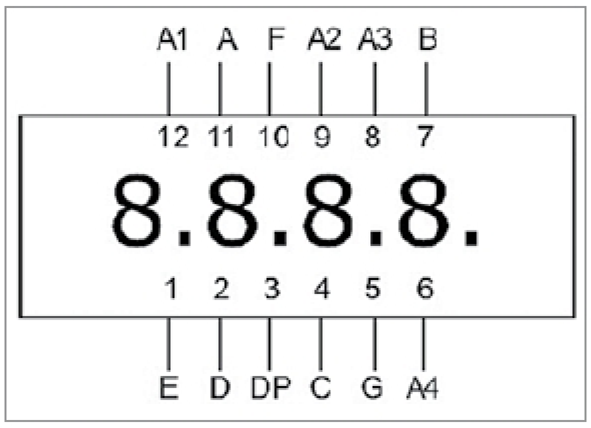

Fig. 6: Pinout of the four-digit common-anode seven-segment display

PARTS LIST

Semiconductors:	
IC1	- LM317 voltage regulator
IC2	- MSP430G2553 MCU
IC3	- HEF4051B multiplexer/ demultiplexer
D1, D2	- 1N4007 rectifier diode
DIS1	- 12-pin, 4-digit, 7-segment display (SUNR056CA)
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 180-ohm
R2	- 100-ohm
Capacitors:	
C1	- 0.1µF ceramic disk
C2	- 10µF 50V electrolytic
C3	- 1µF 63V electrolytic
Miscellaneous:	
BATT.1	- 6V battery
CON1	- 6-pin connector for Bluetooth module
CON2	- 5-pin connector for MAX6675 module
	- HC-05 Bluetooth module
	- MAX6675 module

(All these are available from kitsnspares.com)

10. Connect MSP430 LaunchPad Kit (with MSP430G2553) with the PC using USB type A male to USB type B mini cable.

11. To burn the code to the MCU, select Project→Download→Download active application. After burning is complete, remove the MCU and place it on the temperature measurement target board. It is always good to save the file by giving a filename and clicking on Save.

Troubleshooting. In case there is a problem in running the project, troubleshoot using the steps given below.

1. Download and install MSPDRIVERLIB from *http://www.ti.com/tool/MSPDRIVERLIB*

2. Select Category→FET Debugger →Connection→Elprotonic USB-FPA, Manual Selection→Spy-Bi-Wire.

The firmware performs the following functions:

1. Halt watch-dog timer and initialise the internal clock to 16MHz (digital-controlled oscillator).

2. Initialise USCI for SPI and UART.

3. Initialise the timer to create periodic interrupts.

4. For every timer interrupt, read sensor data from MAX6675 and convert that digital data to temperature.

5. Update the value on seven-segment display.

6. If UART receives ST in ASCII, it will keep sending temperature data to HC-05 module for every timer interrupt.

EFY Note
The source code of this project is available for free download at *source.efymag.com*

Construction and testing

While constructing the circuit, care must be taken to check all components prior to assembly. This includes checking HEF4051B by connecting digital select pins, ground pin, inhibit pin to 0V and V_{DD} pin to 3.6V.

Now, give input to pin Z; it must be reflected at output pin Y0. If Z is high, then Y0 is high, and *vice-versa*. Connect inhibit pin ($\overline{E}$) of HEF4051B to high (V_{DD}). Output Y0 should be floating.

Test the four-digit, seven-segment display (DIS1) using a multimeter in continuity mode, and map the pins including anode and cathode. Pin details of DIS1 used in the project are shown in Fig. 6.

Need for the resistor to limit current to DIS1 is eliminated due to the presence of internal resistance of HEF4051B.

Whenever the system is powered on, the MCU waits for timer interrupt (~1sec). Upon timer interrupt, the MCU starts to acquire temperature data from the thermocouple sensor through MAX6675 module and converts digital data to valid temperature data, and displays the same on the four-digit, seven-segment display (DIS1). EFY


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